

Unit-2

Information Security

- Computer Security- collection of tools design to protect and to hackers
- Network Security- protect data during their transmission
- Internet Security- protect data during their transmission over a collection of interne connected network.
- Information security- is not only about securing information from unauthorized access.

Security Goals



Types of Attacaks

- Dos and Ddos attack
- Man-in-the-middle attack
- Phishing attack
- Password attack
- SQL injection attack
- Cross-site scripting attack

E-commerce security

- Confidentiality
- Integrity
- Availability
- Authenticity
- Encryption

Secure socket layer(SSL)

Secure Hypertext Transfer Protocol(SHTTP)

- **Computer Forensics**

- What is computer forensics?

Forensics is the process of using scientific knowledge for collecting, analyzing, and presenting evidence to the courts. Forensics deals primarily with the recovery and analysis of latent evidence. Latent evidence can take many forms, from fingerprints left on a window to DNA evidence recovered from blood stains to the files on a hard drive. computer forensics as the discipline that combines elements of law and computer science to collect and analyze data from computer systems, networks, wireless communications, and storage devices.

- Why is Computer Forensics Important?
- computer forensics will help you ensure the overall integrity and survivability of your network infrastructure. You can help your organization if you consider computer forensics as a new basic element in what is known as a “defense-in-depth”. To understanding the legal and technical aspects of computer forensics will help you capture vital information if your network is compromised and will help you prosecute the case if the intruder is caught.

- **Two basic types of data are collected in computer forensics.**
- Persistent data is the data that is stored on a local hard drive and is preserved when the computer is turned off.
- Volatile data is any data that is stored in memory, or exists in transit, that will be lost when the computer loses power or is turned off. Volatile data resides in registries, cache, and random access memory (RAM). Since volatile data is ephemeral, it is essential an investigator knows reliable ways to capture it

Unit-3

Security Threats and Vulnerabilities

Threats

- The people eager, willing and qualified to take an advantage of each security vulnerability, and continually search for new exploits and weaknesses.
- Threats can be many like software attacks, theft of intellectual property, identity theft, theft of information and information extortion.
- **Software attacks** means attack by Viruses, Worms, Trojan Horses etc.
- **Malware** means malicious software that is program code or a malicious operations on system.

- **Four main classes of threats:**

- **Structured Threats:** technical person who is trying to access to your network
- **Unstructured Threats** non-technical person trying to access to your network
- **Internal Threats** form inside your network
- **External Threats** from outside your network

- **Malware can be divided in 2 categories:**

- **Infection Methods:**

- **Virus** replicate by hooking them to the program on the host computer. e.g. File Virus, Macro Virus, Boot sector Virus.
 - **Worms** are self replicating but it don't hook to the program on host computer. e.g. slowing down the computer.
 - **Trojan** their purpose is executed their task of either stealing information or steal your valuable data without your knowledge and permission. e.g. FTP Trojans, Proxy Trojans etc.
 - **Bots** they are automated processes that are designed to interact over the internet without the need of human interaction.

- **Malware can be divided in 2 categories:**
 - **Malware Actions:**
 - **Adware** they display ads on computer's or inside individual programs. They monitor your interests and display relevant ads. An attacker can embed malicious code inside the software and adware can monitor user's system activities.
 - **Spyware** is a program or software that monitors user's activities on computer. e.g. Keylogger
 - **Ransomware** is encrypt user's file or will lock your computer. e.g. ransom in exchange.
 - **Scareware** a tool or software is executed it will infect your system or completely destroy.

- **Theft of intellectual property** means copyrights, patents etc.
- **Identity theft** like accessing the computer or social media account of a person by login into the account. e.g. login credentials.

- **Some of these are new generation threats**
 - **Technology with weak security** like very few devices are fully secured and follows information security principles.
 - **Social media attacks** it identify that to steal persons information's.
 - **Mobile Malware** e.g. user to download a game or any unknown application from internet then unintentionally user will install malware or virus in the device.
 - **Outdated security software** updation in security software.
 - **Social engineering**

- **Cyber crime – Mobile security Threats**

- **Types of mobile security Threats:**

- **Web-Based Threats** happen when people visit sites
 - **Phishing through links** are sent through messages, emails, or any social media platforms.
 - **Forced Downloads** drive-by downloads
 - **Physical Threats** physically tries to access your device
 - **No Password Protection**
 - **Encryption**
 - **Network-Based Threats** cybercriminal can steal unencrypted data while people use public WiFi network
 - **Public WiFi** in public provided with public open WiFi
 - **Network Exploits** weakness in the OS in user's mobile device

- **Types of mobile security Threats:**
 - **Application-Based Threats** designed to carry malicious activities.
 - **Malware** unwanted messages to recipients
 - **Spyware** software used to collect specific information

- **Steps to prevent from Mobile Security Threats**

- Application it should be encrypt data transfer
- Update your device software regularly
- Unique passwords
- Do not allow forced downloads from browser
- Stop sharing of network
- Do not add or upload your data to public servers

Hacking Techniques

• **Types of Hackers**

- **White Hat Hackers** who is authorized or the certified hackers who work for the government and organizations .
- **Black Hat Hackers** who is trying to unauthorized access of your system or data
- **Gray Hat Hackers** are not legally authorized hackers, they work for both good and bad intentions

- **Some common Hacking Techniques**

- **Bait and Switch** through advertisement on website's page, after clicking it redirected to malicious web-page.
- **Virus, Trojan and other Spyware** malicious code that installs it on the victim's computer
- **Cookie Theft** different websites store cookies in your browser, that includes our search history, some account passwords.
- **Denial of Service** flooding the network with huge amount of data packets to system down.
- **Keylogger** these log files can contain some useful data that may be account information and different password.

Insecure Network Connections

- An unsecure network most often refers to a free Wi-Fi network, like at a coffeehouse or retail store. It means there's no special login or screening process to get on the network, which means you and anyone else can use it means the Wi-Fi is open to anyone.

How to Secure Network Connections

- Monitor Firewall performance
- Update passwords
- Anti-Virus Software
- Through VPN